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Applied Time Series Analysis
STA457H5S, Winter 2022
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Yule-Walker Equations and Computing the ψ -weights

February 18, 2022

Problem 1. Consider the AR(2) process $\{X_t\}$ satisfying

$$X_t - \phi X_{t-1} - \phi^2 X_{t-2} = W_t$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

a) For what values of ϕ is this a causal process?

b) The following sample moments were computed after observing $X_1, X_2, \dots, X_{200}$:

$$\hat{\gamma}_0 = 6.06, \quad \hat{\rho}_1 = 0.678.$$

Find estimates of ϕ and σ_w^2 by solving the Yule-Walker equations. (If you find more than one solution, choose the one that is causal.)

Solution.

a) The AR characteristic polynomial is derived as being

$$\phi(z) = -\phi^2 z^2 - \phi z + 1$$

So, putting this equal to zero implies

$$\phi(z) = -\phi^2 z^2 - \phi z + 1 = 0 \implies z_1 = -\frac{1 - \sqrt{5}}{2\phi} \quad \text{and} \quad z_2 = -\frac{1 + \sqrt{5}}{2\phi}$$

We can then obtain

$$\begin{cases} |z_1| > 1 \iff |\phi| < \frac{\sqrt{5} - 1}{2} = 0.618 \\ |z_2| > 1 \iff |\phi| < \frac{1 + \sqrt{5}}{2} = 1.618 \end{cases}$$

Hence, the process is causal if

$$|\phi| < 0.618$$

b) The Yule-Walker equations are:

$$\gamma_k - \phi\gamma_{k-1} - \phi^2\gamma_{k-2} = \begin{cases} \sigma_w^2 & k = 0, \\ 0 & k \geq 1. \end{cases}$$

Rewriting the first 3 equations and using $\gamma_k = \gamma_{-k}$ gives

$$\begin{cases} \gamma_0 - \phi\gamma_1 - \phi^2\gamma_2 = \sigma_w^2 \\ \gamma_1 - \phi\gamma_0 - \phi^2\gamma_1 = 0 \\ \gamma_2 - \phi\gamma_1 - \phi^2\gamma_0 = 0 \end{cases}$$

By noting that

$$\gamma_1 = \gamma_0 \rho_1,$$

we then solve the second equation to obtain

$$\phi = \frac{\gamma_0 \pm \sqrt{\gamma_0^2 + 4\gamma_1^2}}{-2\gamma_1} = -\frac{1}{2\rho_1} \pm \sqrt{\frac{1}{4\rho_1^2} + 1}$$

Subsequently, inserting the estimated value $\hat{\rho}_1 = 0.678$ in the equation above gives the solutions

$$\hat{\phi} = \begin{cases} 0.505 \\ -1.979 \end{cases}$$

and we choose the causal solution given by

$$\hat{\phi} = 0.505$$

In order to estimate σ_w^2 , it would suffice to multiply the third equation of the Yule-Walker set of equations by ϕ^2 and add it up to the first equation to derive

$$-\phi^3\gamma_1 - \phi\gamma_1 - \phi^4\gamma_0 + \gamma_0 = \sigma_w^2$$

Hence,

$$\hat{\sigma}_w^2 = -\hat{\phi}^3\hat{\gamma}_1 - \hat{\phi}\hat{\gamma}_1 - \hat{\phi}^4\hat{\gamma}_0 + \hat{\gamma}_0 = 3.062$$

where an estimate for γ_1 is obtained using the following relation

$$\hat{\gamma}_1 = \hat{\gamma}_0 \hat{\rho}_1 = 4.109$$

Problem 2. Identify the following models as ARMA(p,q) models (watch out for parameter redundancy), and determine whether they are causal and/or invertible. In each case $\{W_t\} \sim wn(0, 1)$. If any model is shown to be causal, then further compute its corresponding first five coefficients $\psi_0, \psi_1, \psi_2, \psi_3, \psi_4$ in the causal linear process representation $X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}$.

a) $X_t = 0.8X_{t-1} - 0.15X_{t-2} + W_t - 0.3W_{t-1}$

b) $X_t = X_{t-1} - 0.5X_{t-2} + W_t - W_{t-1}$

c) $X_t - \frac{9}{4}X_{t-1} - \frac{9}{4}X_{t-2} = W_t - 3W_{t-1} + \frac{1}{9}W_{t-2} - \frac{1}{3}W_{t-3}$

Solution.

a) The AR and MA characteristic polynomials are derived as follows:

$$\phi(B) = 1 - 0.8B + 0.15B^2$$

$$\theta(B) = 1 - 0.3B$$

The first step in specifying the order of the ARMA process is to check whether the given polynomials have a common factor (checking for *parameter redundancy*). If that is the case, then we need to remove the common factor from both polynomials, and only after that our specification of the model order will be correct.

Basically, two polynomials would have a common factor if and only if they share a common root. So, let us first compute the roots of the given polynomials, i.e.,

$$\phi(B) = 1 - 0.8B + 0.15B^2 = 0 \implies B_1 = 2 \quad \text{and} \quad B_2 = \frac{10}{3}$$

$$\theta(B) = 1 - 0.3B = 0 \implies B = \frac{10}{3}$$

As it is seen, the given polynomials share a common root $B = \frac{10}{3}$, which indicates that they have a common factor. Since $\theta(B)$ is a first-order polynomial in this specific problem, we can easily verify that the common factor between $\phi(B)$ and $\theta(B)$ should be $(1 - 0.3B)$.

Now, since we are interested to write down the given ARMA model in the form of

$$\phi(B)X_t = \theta(B)W_t$$

we would need to factorize the second-order polynomial $\phi(B)$ into multiplication of two first-order polynomials, one of which is $(1 - 0.3B)$.

A naive approach to find the second factor of $\phi(B)$ would be to perform the long division for polynomials of the following fraction

$$\frac{1 - 0.8B + 0.15B^2}{1 - 0.3B}$$

whose remainder will clearly be zero.

However, an easier, and more elegant, approach to find the second factor of $\phi(B)$ would be to resort to the applications of the *Fundamental Theorem of Algebra*. The fundamental theorem of algebra states that every polynomial of order n , i.e.,

$$p(x) = 1 - c_1x - c_2x^2 - \dots - c_nx^n$$

can be written in the form

$$p(x) = (1 - \frac{x}{\lambda_1})(1 - \frac{x}{\lambda_2}) \dots (1 - \frac{x}{\lambda_n})$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the n roots of the polynomial $p(x)$, which may be real- or complex-valued.

Using the fundamental theorem of algebra, we can then factorize $\phi(B) = 1 - 0.8B + 0.15B^2$ as follows:

$$\phi(B) = (1 - \frac{B}{2})(1 - \frac{B}{10/3}) = (1 - 0.5B)(1 - 0.3B)$$

Now, if write the ARMA model in terms of the characteristic polynomials, e.g., $\phi(B)X_t = \theta(B)W_t$, we obtain

$$(1 - 0.5B)(1 - 0.3B)X_t = (1 - 0.3B)W_t$$

which would simplify to

$$(1 - 0.5B)X_t = W_t$$

Hence, the actual (reduced) characteristic polynomials of the given ARMA model would become

$$\phi_{\text{red}}(B) = 1 - 0.5B$$

$$\theta_{\text{red}}(B) = 1$$

which implies that the given model is ARMA(1,0), or simply an AR(1) process.

Since, this is pure AR model, it is always invertible (since they contain no MA terms). Furthermore, we have that

$$\phi_{\text{red}}(B) = 1 - 0.5B = 0 \implies B = 2$$

which falls outside the unit circle in the complex plane. Hence, the model is causal.

Subsequently, we can write the model in the causal linear process representation by obtaining the ψ weights as follows:

$$\psi(B) = \frac{\theta_{\text{red}}(B)}{\phi_{\text{red}}(B)} = \frac{1}{1 - 0.5B} = 1 + 0.5B + 0.5^2B^2 + 0.5^3B^3 + 0.5^4B^4 + \dots$$

which implies that

$$\psi_0 = 1 \quad \text{and} \quad \psi_i = 0.5^i \quad \text{for } i = 1, 2, \dots$$

Hence,

$$\psi_0 = 1, \psi_1 = 0.5, \psi_2 = 0.5^2, \psi_3 = 0.5^3, \psi_4 = 0.5^4$$

b) The AR and MA characteristic polynomials are derived as follows:

$$\phi(B) = 1 - B + 0.5B^2$$

$$\theta(B) = 1 - B$$

The first step in model specification would be to check for parameter redundancy. For this purpose, we

obtain the roots for each of the characteristic polynomials given above, i.e.,

$$\phi(B) = 1 - B + 0.5B^2 = 0 \implies B_1 = 1 + i \quad \text{and} \quad B_2 = 1 - i$$

$$\theta(B) = 1 - B \implies B = 1$$

where $i = \sqrt{-1}$ is the *imaginary number*. It is clear that two polynomials do not have a common root, so no common factor exists between them. Accordingly, the order of the model can be specified as being ARMA(2,1).

Furthermore, the model is not invertible since the root of the MA polynomial ($B = 1$) falls on the unit circle. However, the model is causal as we have

$$|B_1| = |1 + i| = \sqrt{1 + 1} = \sqrt{2} > 1$$

$$|B_2| = |1 - i| = \sqrt{1 + 1} = \sqrt{2} > 1$$

In order to obtain the causal linear process representation of the model, we can use either of the following methods:

Method 1. Note that the sequence $\{\psi_j\}$ is determined by the relation $\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\theta(B)}{\phi(B)}$, or by the identity

$$(1 - \phi_1 B - \dots - \phi_p B^p)(\psi_0 + \psi_1 B + \dots) = 1 + \theta_1 B + \dots + \theta_q B^q$$

Equating coefficients of $z^j, j = 0, 1, \dots$ we find $1 = \psi_0$

$$1 = \psi_0$$

$$\theta_1 = \psi_1 - \psi_0 \phi_1$$

$$\theta_2 = \psi_2 - \psi_1 \phi_1 - \psi_0 \phi_2$$

or equivalently,

$$\psi_j - \sum_{k=1}^p \phi_k \psi_{j-k} = \theta_j, \quad j = 0, 1, 2, \dots$$

where

$$\theta_0 := 1 \quad \text{and} \quad \theta_j := 0 \text{ for } j > q \quad \text{and} \quad \psi_j := 0 \text{ for } j < 0$$

Then, we may write:

$$\begin{cases} \psi_0 = 1 \\ \psi_1 - \psi_0\phi_1 = \theta_1 \\ \psi_2 - \psi_1\phi_1 - \psi_0\phi_2 = \theta_2 \\ \psi_3 - \psi_2\phi_1 - \psi_1\phi_2 - \psi_0\phi_3 = \theta_3 \\ \psi_4 - \psi_3\phi_1 - \psi_2\phi_2 - \psi_1\phi_3 - \psi_0\phi_4 = \theta_4 \end{cases}$$

Next, we substitute the values for the parameters existing in the set of equations given above using

$$\phi_1 = 1, \phi_2 = -0.5, \phi_3 = 0, \phi_4 = 0$$

$$\theta_1 = -1, \theta_2 = 0, \theta_3 = 0, \theta_4 = 0$$

Subsequently, we solve the equations recursively, which yields

$$\psi_0 = 1, \psi_1 = 0, \psi_2 = -0.5, \psi_3 = -0.5, \psi_4 = -0.25$$

Method 2. We have that

$$\psi(B) = \frac{\theta(B)}{\phi(B)} = \frac{1-B}{1-B+0.5B^2} = (1-B)\left(\frac{1}{1-B+0.5B^2}\right)$$

Then, the second expression can be written by its associated Maclaurin series as follows:

$$\frac{1}{1-B+0.5B^2} = 1 + B + \frac{B^2}{2} - \frac{B^4}{4} + \dots$$

Subsequently, we would have

$$\psi(B) = (1-B)\left(\frac{1}{1-B+0.5B^2}\right) = (1-B)\left(1+B+\frac{B^2}{2}-\frac{B^4}{4}+\dots\right) = 1-0.5B^2-0.5B^3-0.25B^4+\dots$$

Hence,

$$\psi_0 = 1, \psi_1 = 0, \psi_2 = -0.5, \psi_3 = -0.5, \psi_4 = -0.25$$

c) The AR and MA characteristic polynomials are derived as follows:

$$\phi(B) = 1 - \frac{9}{4}B - \frac{9}{4}B^2$$

$$\theta(B) = 1 - 3B + \frac{1}{9}B^2 - \frac{1}{3}B^3$$

The first step in model specification would be to check for parameter redundancy. It is not difficult to verify that these polynomials can be factorized as follows:

$$\phi(B) = \frac{1}{4}(1 - 3B)(3B + 4)$$

$$\theta(B) = \frac{1}{9}(1 - 3B)(B^2 + 9)$$

Since both polynomials have a common factor $(1 - 3B)$, we have to remove this factor from the AR and MA polynomials before specifying the order of the ARMA model. That is,

$$\phi_{\text{red}}(B) = \frac{1}{4}(3B + 4) = 1 + \frac{3}{4}B$$

$$\theta_{\text{red}}(B) = \frac{1}{9}(B^2 + 9) = 1 + \frac{1}{9}B^2$$

So, this is an ARMA(1,2) process. Also, this model is causal since

$$\phi_{\text{red}}(B) = 1 + \frac{3}{4}B = 0 \implies B = -\frac{4}{3} \implies |B| > 1$$

and it is invertible as

$$\theta_{\text{red}}(B) = 1 + \frac{1}{9}B^2 = 0 \implies B_1 = 3i \quad \text{and} \quad B_2 = -3i \implies |B_1| = |B_2| > 1$$

In order to derive the causal linear process representation of the model, we can employ any of the methods described in parts (a) and (b), i.e., let us write down the Maclaurin series for the following function

$$\frac{1}{1 + \frac{3}{4}B} = 1 - \frac{3}{4}B + \frac{9}{16}B^2 - \frac{27}{64}B^3 + \frac{81}{256}B^4 + \dots$$

Then, we can obtain the ψ weights as follows:

$$\psi(B) = \frac{\theta_{\text{red}}(B)}{\phi_{\text{red}}(B)} = (1 + \frac{1}{9}B^2)(1 - \frac{3}{4}B + \frac{9}{16}B^2 - \frac{27}{64}B^3 + \frac{81}{256}B^4 + \dots) = 1 - \frac{3}{4}B + \frac{97}{144}B^2 - \frac{97}{192}B^3 + \frac{97}{256}B^4 + \dots$$

Hence,

$$\psi_0 = 1, \psi_1 = -\frac{3}{4}, \psi_2 = \frac{97}{144}, \psi_3 = -\frac{97}{192}, \psi_4 = \frac{97}{256}$$

or

$$\psi_0 = 1, \psi_1 = -0.75, \psi_2 = 0.674, \psi_3 = -0.505, \psi_4 = 0.379$$